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Role of sports psychology and sports nutrition in return to play from musculoskeletal injuries in professional soccer: an interdisciplinary approach

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Abstract
Musculoskeletal injuries are prevalent in professional soccer and can result in lost training time or match play. It is intuitive that the "return to play" (RTP) pathway will depend, in large part, on the expertise of sports medicine practitioners (e.g. surgeons, physicians, physiotherapists) responsible for player's recovery. Consensus statements on returning athletes to sport following injury acknowledge the contributions of sport psychology and sports nutrition. However, specific consideration on how to integrate these two recognized – but often overlooked components of injury rehabilitation – into existing sport medicine approaches has yet to be examined. Using a framework of milestones directed by the medical physician and physical trainer, the evidence is summarized and suggestions provided on the integration of sports psychology and sports nutrition into an interdisciplinary RTP approach. We examine recovery from a phase approach (acute injury and functional recovery) to highlight interdisciplinary opportunities in the management of musculoskeletal soccer injuries. An interdisciplinary approach is understood to achieve outcomes that could not be achieved within the framework of a single discipline. The incorporation of sports psychology and nutrition theoretically compliment milestones used in current medically-based RTP models. Our hope is that this article serves as a catalyst for interdisciplinary practice and research – not only in sports nutrition and sports psychology – but across all sport and exercise disciplines.

Keywords: *Injury, muscle, medicine, exercise, psychology, nutrition*

Highlights
- A musculoskeletal injury to a player presents an opportunity for both the sport nutritionist and sports psychologist to engage with the player, as well as the sports medicine department.
- The interdisciplinary theme is important as it *integrates* concepts, theories, methods, and techniques from two or more disciplines to achieve outcomes (including new approaches) that could not be achieved within the framework of a single discipline.
- The incorporation of sports psychology and nutrition theoretically compliment milestones used in current medically–based return to play models. However, it remains uncertain whether or how adoption of an interdisciplinary approach facilitates a more effective and/or comprehensive recovery in comparison to single discipline approaches.
- Providing the players with appropriate sports nutrition and psychological "tools" to manage the injury process can also extend and remain relevant to when players are back in full training and competition.

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Introduction

An injury may affect any player who participates in soccer (football) either for health and enjoyment or competitive/professional purposes (Wall, Morton, & van Loon, 2015). In soccer, an injury may be defined as any physical complaint sustained by a player, resulting from a match or training, irrespective of the need for medical attention ("medical attention injury") or time loss from training or match play ("time loss injury") (Ardern et al., 2016; Fuller et al., 2006). The process and duration required to return a player to training and competition following a "time loss" injury is complex and will depend on the injury classification (Rae & Orchard, 2007). As such the return to play (RTP) process should be regarded as a continuum, paralleled with recovery and rehabilitation (Ardern et al., 2016). This continuum has been proposed to have three main stages: return to participation (athlete is physically active, but not yet medically, physically or psychologically ready to return to sport), return to sport (athlete has returned to defined sport but is not yet performing at the desired level) and finally, return to performance. Return to performance (play, in team sports, RTP) can be defined as the player performing at or above their pre-injury level (Ardern et al., 2016).

The present manuscript will focus on the value of an interdisciplinary approach in treating muscular-skeletal injuries, given that they account for up to 46% of all injuries in professional soccer (Ekstrand, Hagglund, & Walden, 2011). The majority (~90%) of musculoskeletal injuries affect the major muscle groups of the lower limbs. Acute hamstring injury has been reported to be the most common non-contact musculoskeletal injury in sports involving high-speed running with a high risk of reinjury (De Vos et al., 2014). Musculoskeletal injuries are caused by trauma, either from direct contact (heavy extrinsic compressive force) from an opposing player or indirectly by a muscle strain (myofibers are exposed to an excessive intrinsic tensile force). Musculoskeletal injuries result in pain, which can impact players ability to train and compete. Approximately 35% of soccer players miss training or matches due to musculoskeletal injuries each season (Ekstrand et al., 2011). Half of musculoskeletal injuries will be minor, resulting in absences of less than a week. However, more severe musculoskeletal injuries may result in player absences of 11 days up to 50 weeks. In the English Premier League, the financial cost of 32 days lost due to a hamstring injury has been calculated at £209,000 (JTL, S.I.I, 2017). Therefore, speeding the players return from musculoskeletal injures has both practical and financial implications. It is also important to ensure that players' return in a time and manner that prioritizes their health and well-being in addition to concerns regarding expedience.

Efforts to ensure that returning players are physically and psychologically ready to do so are best guided by an interdisciplinary approach. An interdisciplinary approach is one in which the expertise of scholars and practitioners from different disciplines are brought to bear on the research or clinical problem in question. This approach is understood to achieve outcomes (including new approaches) that could not be achieved within the framework of a single discipline (Piggott, Müller, Chivers, Papaluca, & Hoyne, 2019). It is intuitive that the approach to how a player RTP will depend, in part, on the expertise of the individual(s) responsible for their recovery. Physicians and physiotherapists play a major role in assisting the player and directing the RTP process following injury or surgery (McCall, Lewin, O'Driscoll, Witvrouw, & Ardern, 2017). Consensus statements on returning athletes to sport following injury recognize the important contributions of other disciplines, including sports science, coaching, and strength and conditioning (Ardern et al., 2016). Indeed, within Ardern et al.'s consensus statement, it is recognized that biopsychosocial factors are important in the RTP process. However, specific guidance on how best to integrate sports psychology and sports nutrition perspectives and strategies into existing medical approaches to sport injury rehabilitation and RTP is needed.

To this end, the purpose of this paper is to examine the different phases of recovery from musculoskeletal injury, with an eye towards articulating where and how sport psychology and sports nutrition can be integrated into the RTP process. Research evidence should guide RTP recommendations. However, it has been acknowledged that practical experience is also required, in combination with knowledge of the injury, the sport and the individual player (McCall et al., 2017). Therefore, the personal perspective and experience of sports psychologists and sports nutritionists will be combined with the available research in those areas. The phases of RTP will be differentiated as *acute injury* and *functional recovery*. Sports nutrition and sports psychology integration will be discussed in the context of the milestones established by the club's medical physician and physiotherapist (in partnership with the physical trainer). Recommendations are relevant to both male and female players.

An interdisicplinary approach

A consensus statement on return to sport has previously provided a framework for RTP decisions

based on the strategic assessment of risk and risk tolerance (StARRT; readers are referred to Ardern et al., (2016)). It is important to first identify a leader who will establish a clear direction, while listening and providing supervision to the team members during the RTP process. This leader will be heavily conditioned by the cultural background of the team in question, league and/or sport. In a process that starts with an injury, we concur with previous consensus statements that the leader should be an experienced medically-trained individual with a daily presence in team activities (Ardern et al., 2016). This leader should have a mastery of communication strategies and an ability to foster collaborative decision-making. Given that all medical practitioners typically report to the team physician, the latter may be ideally positioned to fulfil a leadership role. Determining the composition and roles within a decision-making team should be made early in the RTP process (Ardern et al., 2016) (milestone 1). Decision-making efforts should utilize a player-centered approach. A player-centered approach is characterized by player empowerment and engagement (i.e. education about the injury and player involvement in the planning of their rehabilitation programmes); ongoing feedback about progress towards players' rehabilitation goals; and transparent, honest and frequent communication between the player and treatment team members (King, Roberts, Hard & Ardern, 2019). Guided by a player-centered approach, sport-medicine practitioners attempt to foster player autonomy (via rationales for rehabilitation programme decisions, input in decisions, taking athletes' perspectives, giving athletes choices and options) and ensure that players voices, perspectives and experiences remain at the forefront of decision-making processes (Ahmed, Defoe, West, & Blake, 2018; Podlog & Eklund, 2020). It is also important to note that decision-making efforts should ideally occur within a culture of trust, consensus, and one supporting training and development across all disciplines (Ahmed et al., 2018; King, Roberts, Hard, & Ardern, 2019). Such an environment will facilitate a path toward interdisciplinarity (Van Raalte, Vincent, & Dickens, 2019). This approach may foster the players confidence that they are receiving comprehensive care, from qualified health and sport medicine practitioners. Such care may increase player well-being. However, while the benefits of an interdisciplinary approach seem intuitive, further empirical work is needed to document the value of such an approach for balancing well-being considerations with performance ones (Heil & Podlog, 2012).

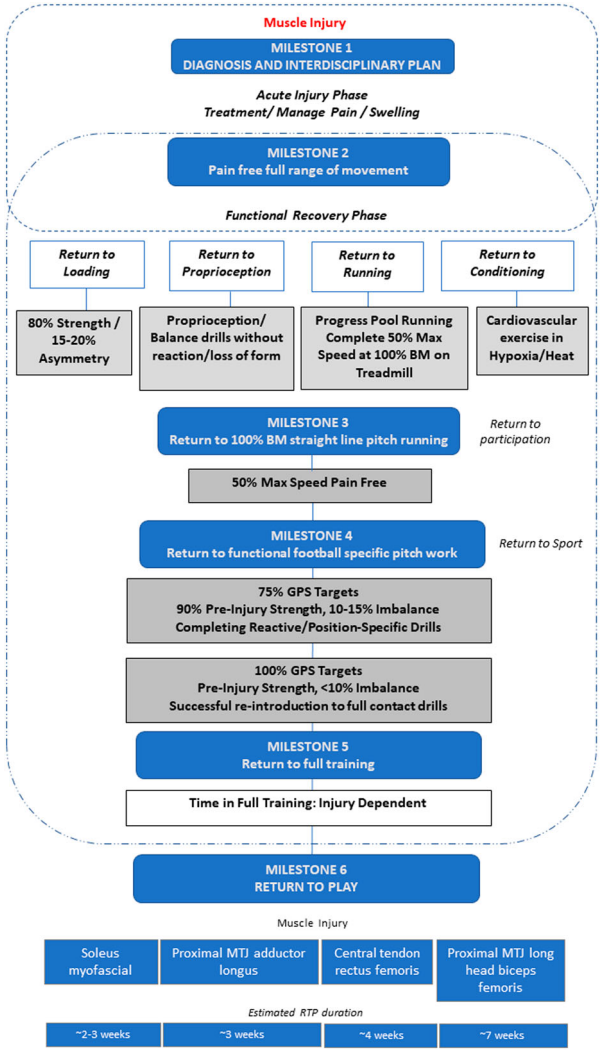

Figure 1. Schematic of the key milestones in the return to play process. Shaded "grey" boxes included a guide to criteria for milestone progression. Injury time lines are estimations only based on FC Barcelona data and clinical experience that do not consider severity of injury, player specific or football specific factors. MTJ = Musculotendinous junction.

Acute injury phase

The acute injury phase is also referred to as the immobilization phase or injury management phase (Ardern et al., 2016). It can be defined from the moment the injury occurs to the commencement of active mobilization of the injured area (return to participation). The duration of this phase, conditional upon the severity of the injury, will range from hours to several days. During this time, an initial meeting of the interdisciplinary team should take place to ensure communication among all relevant stakeholders (Figure 1).

An accurate diagnosis is the first step in the RTP process, while an initial prediction of the RTP time-

line (considering the minimum risk for re-injury) is often the first question asked by the player and coaching staff. Articulating a path and timeline for RTP may be particularly important for alleviating player, coach and management apprehensions about time-loss concerns. Physiotherapists, sport psychologists and nutritionists can collaborate to implement best-practice goal-setting guidelines that incorporate healing and recovery progressions into a rehabilitation programme that helps players gain a sense of the RTP time-line. The nutrition plan should complement the exercise progressions. Doing so may be invaluable in helping players feel a greater sense of autonomy – that is, a sense of control – over their RTP (Podlog, Heil, & Schulte, 2014).

During the acute phase, psychological issues commonly include pain, uncertainties regarding the nature and extent of injury damage, and negative thoughts and emotions (Heil & Podlog, 2012). Pain can take a variety of forms (e.g. physical sensations associated with bodily trauma; emotional pain associated with the realization of loss of a valued activity) and is defined as a sensory and emotional experience associated with actual or potential tissue damage, or described in terms of such damage and its impact (Merskey & Bogduk, 1994). Catastrophic thinking ("I'll never come back from this"), negative thoughts ("Why me", "Why now?") and emotions (frustration, anger, hopelessness) may exacerbate the perceptions of pain. Such thinking contributes to the feeling that one cannot cope with the pain, is associated with augmented psychological distress, and compromises function (Jones, Rollman, White, Hill, & Brooke, 2003). The sports medicine department (provided with appropriate training) can aim to mitigate pain responses and corresponding feelings of distress through techniques such as muscular relaxation (with non-affected limbs/muscles), diaphragmatic breathing, and potentially through the use of imagery (Podlog, Heil, & Podlog, 2018). The content of imagery sessions can be determined based on communication between interdisciplinary practitioners and the injured player. For example, nutritionists, physiotherapists, physicians and sport psychologists could develop injury-specific imagery scripts that incorporate descriptions of anatomical, physiological or healing processes (e.g. ingested protein being used to build new muscle) along with key breathing and relaxation principles (sport psychologist).

The priority in this phase is to respect the principles of protection, optimal loading, ice, compression and elevation (POLICE) of the affected limb (Bleakley, Glasgow, & MacAuley, 2012). Recommendations in the acute phase are to prevent further retraction of the ruptured muscle, minimize the size of the haematoma and reduce the size of the connective tissue scar. Care must be taken when altering the inflammatory response. Specifically, the ingestion of non-steroidal anti-inflammatory drugs may inhibit strong signals that promote the regenerative process (Jarvinen, Jarvinen, & Kalimo, 2013).

For players intent on returning to the pitch as quickly as possible, respecting a period of restricted or complete inactivity may be particularly challenging from a psychological standpoint. Sport psychologists and sport medicine providers can work collaboratively to help alleviate player distress regarding physical incapacitations and to provide education and clear and consistent communication regarding the importance of physical inactivity in healing and recovery progressions (Figure 1). Such efforts are imperative given that uncertainties about injury diagnosis (severity, implications) and doubts regarding the efficacy of the treatment received are significant predictors of treatment adherence. Podlog et al. (2018) provides a guideline for injury education – developed in collaboration between sport psychologists and sport medicine providers – that can be used by players and practitioners to address common uncertainties and reduce feelings of apprehension about one's injury, the prognosis, and the course of treatment. The education guideline serves as a heuristic for helping players build a knowledge base regarding their recovery and to develop a sense of personal investment in the recovery process. Example content areas include basic anatomy of the injured area, active/passive rehabilitation methods, potential problems with pain and coping strategies, and rationales for limits on physical activity during the acute injury phase. A variety of approaches may help players manage negative thoughts and emotions, including cognitive reframing strategies (self-talk); written emotional disclosure; and a focus on internal/controllable factors when commencing the functional recovery phase (Podlog et al., 2018).

During the first 48–72 hours of the acute phase, the physiotherapist may use cold compression therapy, pulse massage (dynamic compression systems to enhance blood flow) and compression bandages. Consistent with the psychology of sport injury research, effective communication between the sports medical team and player can include the provision of rationales for treatment recommendations and decisions, offering players choices and options – when and where feasible – and making an effort to show players that their perspectives are considered and valued. Adopting such strategies has been shown to reinforce player autonomy (control) over the recovery process (Chan et al., 2009). This is a vital consideration during the acute phase of recovery, when players may notoriously feel a lack of control

over their body and the course of their recovery/RTP. For the purpose of this paper, the physical trainer can be defined as the individual(s) responsible for the functional recovery phase of RTP (Figure 1).

During the acute phase, nutritional experts and sport psychologists can also share research with one another and with injured players highlighting the potential impact of injury on maladaptive eating behaviours (e.g. bulimia, anorexia, etc), body dissatisfaction (e.g. negative feelings of self-worth associated with weight gain) and physical healing in high-performance individuals. For example, work by Reel and colleagues (2018) with professional dancers revealed that injury-induced modifications to dancers' nutritional intake (e.g. reduction of calories, control of food intake) to avoid weight gain during periods of increased physical inactivity. Some dancers also highlighted negative thoughts and feelings towards their bodies during injury-periods, indicating concerns over weight-gain, muscle loss, and diminished dance-specific fitness. Sport psychologists and nutritionists can work collaboratively to identify players who may be at-risk for maladaptive eating behaviours and/or body-dissatisfaction; to mitigate and/or treat such issues when they arise; and to provide players with nutrition programmes that ensure sufficient energy for repair and maintenance of lean mass (Reel et al., 2018).

Energy expenditure could be higher during the acute injury phase as a result of the increased energy expenditure of healing combined with the daily early-phase rehabilitation. Thus, caution is advised against dropping total calorie intake below 2750 Kcal during the acute injury phase, with 2750–3250 Kcal seeming an appropriate range, dependent upon the level of immobilization. It has recently been demonstrated, using Doubly Labelled Water, that the daily energy expenditure of an injured Premier League player (ruptured ACL) was 3178 Kcal (Anderson et al., 2019) which is only slightly less than the 3500 Kcal reported in the same team during a 2-match playing week (Anderson et al., 2017). Whilst some reduction of carbohydrate may be required dependent on the level of immobilization, a previous Premier League footballer case study demonstrated that a large reduction in carbohydrate, and total calories, resulted in a substantial loss of total lean body mass (5.8 kg) which took 8 weeks to regain (Milsom, Barreira, Burgess, Iqbal, & Morton, 2014).

It is well accepted, in laboratory based studies, that limb immobilization not only reduces muscle protein synthesis but also induces a degree of anabolic resistance to protein, the latter of which could be attenuated through increased amino acid ingestion (Wall et al., 2015). The risk is that players reduce total protein intake in response to their gym and field-based training loads being reduced. Increasing daily protein intake to 2.3 g/kg body mass may prevent the loss of lean mass during reduced calorie intake. Protein ingestion should be equally distributed during the day with 30–40 g of casein protein ingested pre-sleep (Trommelen et al., 2018). While sport psychologists should not attempt to provide nutritional information, nutrition experts can advise sport psychologists to reinforce messages to players to follow recommended protein intake and other nutritional guidelines discussed below.

To complement the aforementioned protein ingestion, the ingestion of 5 g of fish oil supplements daily (proving 3 g eicosapentaenoic acid and 2 g docosahexaenoic acid) has been shown to prevent muscle atrophy during limb immobilization in active females through increases in myofibrillar muscle protein synthesis (McGlory et al., 2019). It would be suggested that this strategy start immediately following the injury. As 5 g is in excess of quantities achievable in the standard diet, supplementation is warranted with due consideration given to supplement safety and appropriate batch testing for contaminants.

Whilst creatine loading has not been shown to successfully preserve lean muscle mass during immobilization (Backx et al., 2017), it has been shown to improve the recovery of muscle mass following immobilization (Op 't Eijnde, Urso, Richter, Greenhaff, & Hespel, 2001). Therefore, creatine loading (20 g/d for 5 days) may be considered as soon as the injury occurs, via the easy addition of creatine monohydrate to the players morning smoothies or protein shakes.

Low vitamin D concentrations have been shown to attenuate muscle regeneration both *in vitro* and *in vivo*. In terms of muscle regeneration, it would appear that target concentrations of 75 nmol/L should be maintained. Data from the English Premier League suggests that many players in the winter months are below this threshold (Morton et al., 2012) and should, therefore, consider daily supplementation with 2000 iU of vitamin D_3. A practical strategy would be to regularly and appropriately assess vitamin D concentration and aim to achieve 75 nmol/L in all players throughout the season in case a musculoskeletal injury occurs (Owens, Allison, & Close, 2018).

Functional recovery phase

The functional recovery phase may be defined as the structured and deliberate activation of the injured muscle and is also known as return to participation

(Ardern et al., 2016). Once the severity and timelines of the injury are shared with the sports medicine team, a joint programme will be agreed with the primary aim of recovering the physical performance of the player as soon as possible. The functional recovery phase can begin by collating key performance indicators from previous strength and conditioning data and the sports science performance analysis team (Figure 1). These data allow an assessment of the impact of the injury on a players current training status, allows progress to be monitored, and enables milestone targets to be set. The recovery programme before the player returns to "body mass straight line pitch running" will be primarily delivered by a combination of team physician and physiotherapist with supplementary conditioning work managed by the physical trainer, defining the player's "can do's". At each milestone a progress meeting ensures the sports nutrition and sports psychology strategies are integrated within the interdisciplinary framework (Figure 1).

A key issue during the functional phase is the player's ability to summon motivation, particularly in the case of severe/long term injuries (Brewer & Redmond, 2017). As motivation to engage in recommended practitioner guidelines (e.g. nutrition, medications, avoidance of alcohol use, psychological coping skills) may diminish over the course of rehabilitation, ongoing communication between treatment providers and players' is beneficial in mitigating and/or dealing with motivational decrements. Furthermore, motivation to RTP appears to be an important factor discriminating returning versus non-returning players, holding relevance for the quality of post-injury performances (Brewer & Redmond, 2017). Perceptions of athletic competence are often thrown into question during the recovery phase and motivation-related factors are pertinent to confidence restoration following injury (Magyar & Duda, 2000). Uncertainties about one's ability to master challenging rehabilitation exercises, to regain physical form/function, and to achieve pre-injury goals, are prevalent competence-based concerns (Podlog & Eklund, 2020). Moreover, as indicated, autonomy beliefs may be compromised as players feel a loss of control over their bodies, an inability to control the speed of recovery, a lack of influence over their rehabilitation regimens, and internal and external pressures to RTP. Feelings of isolation and alienation from teammates are also common during functional recovery. The reality that one's football participation is temporarily (or indefinitely) suspended may engender a loss of athletic identity and diminish perceptions of self-worth ("football isn't just something I do, it's who I am"). Thus, during the entirety of the RTP process and especially during periods of inactivity, a nutrition plan can provide "structure" to the player's day. Nutrition can be used as one internal/controllable factor that the player can take "ownership" of (Murphy, Foreman, Simpson, Molloy, & Molloy, 1999). During serious injury, players can become disillusioned and switch focus from performance nutrition to eating for comfort, including "binge eating" (Reel et al., 2018). This may be because RTP timelines can impact players thoughts (e.g. self-efficacy beliefs, self-talk) and emotions (anger, frustration, sadness), which in turn influence behaviours (e.g. adherence to rehab programme).

A self-determination theory approach (Ryan & Deci, 2000) may be considered in addressing the psychosocial challenges of functional recovery. Self-determination theory focuses on the 'why' of behaviour and the ways in which motivation quantity and quality is shaped and nourished by players' interaction with their surrounding social environments. In particular, the extent to which players display autonomous (internalized and self-determined) forms of motivation is predicated by the extent to which their basic psychological needs (competence, autonomy, and relatedness) are supported. Competence-enhancing strategies which have demonstrated value in facilitating injury recovery include goal setting, imagery, mastery experiences (e.g. functional progressions), and role modelling (Podlog, Dimmock, & Miller, 2011). Goal-setting interventions will facilitate rehabilitation adherence, while a number of experimental investigations demonstrate the benefit of imagery in enhancing functional mobility, self-efficacy, and pain management (Zach et al., 2018). Similarly, evidence suggests that mastery experiences in the form of functional progressions and the experience of vicarious success (seeing fellow players/role models make an effective recovery) facilitate efficacy beliefs that a return to elite sport participation is possible (Brewer & Redmond, 2017).

Body composition tracking can be used in the goal-setting process through the regular assessment of fat (evaluation of energy balance) and lean muscle mass, making it a valuable tool in the RTP process. Anonymous data may also be shared of body composition changes in individuals who have experienced a similar injury and successfully RTP. The two most common methods include dual-energy X-ray absorptiometry (DXA) and skinfold assessment. If DXA is used, it is possible to assess changes in specific regions, that is, the injured versus non-injured limb. The advantage of skinfold measures, however, is that assessments can be performed more frequently. Regardless of the technique used, body composition management can be used in the goal-setting process as well as providing valuable consultation time to

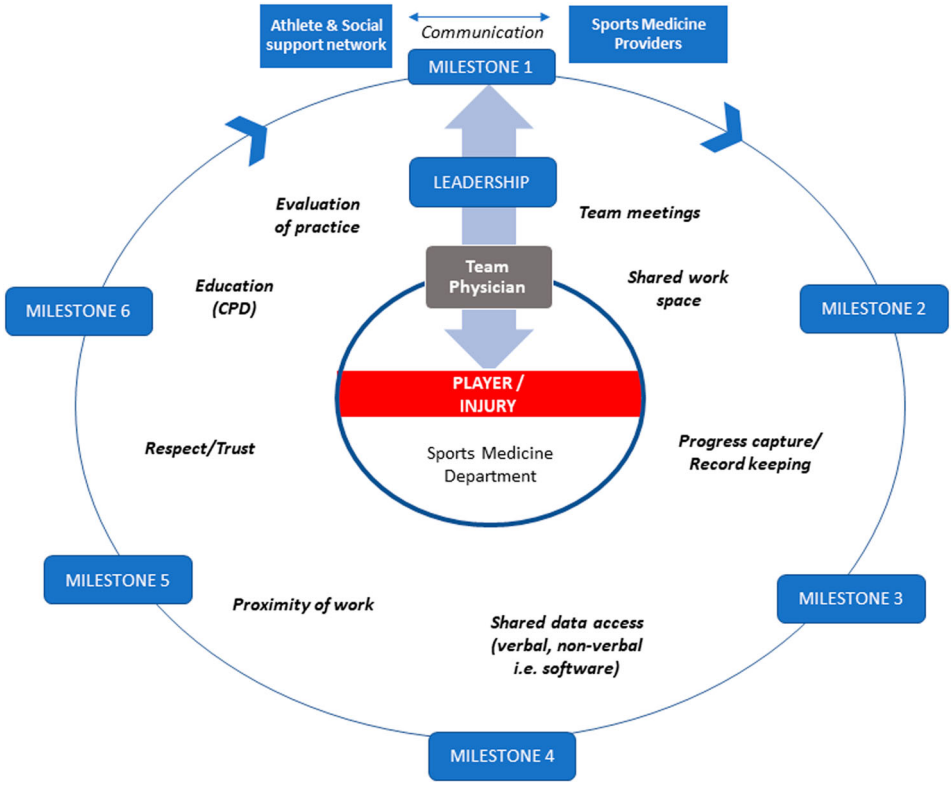

Figure 2. An interdisciplinary framework. Inner circle includes individuals responsible for the players return to play (not limited to those disciplines discussed in the present manuscript). Outer circle provides target milestones as detailed in Figure 1. Within circles are "facilitating factors" to promote interdisciplinary approach. CPD = Career professional development. It is advised to include education opportunities sessions across all disciplines.

understand the strengths and opportunities in the players daily, training and match nutrition strategies (Evans & Hardy, 2002). That said, it is essential that players do not become too focused on preventing any increase in body fat as this may compromise recovery – therefore, the sport nutritionist should frame the data correctly providing appropriate education as a form of social support (Figure 2). Incorporation of body composition testing into goal-setting processes highlights the need for integrative efforts between the nutritionist and sport psychologist.

Relatedness issues may be addressed through various means, including (but not limited to) the provision of social support and the development of positive player-practitioner rapport. Identity issues may be addressed by helping footballers perceive rehabilitation as their "new competitive challenge", one requiring the same set of skills as those needed for effective training and performance. As such, players should be encouraged to take personal responsibility for their training which is orientated towards a task-goal structure. Players should complete exercises with precision, and a willingness to reach for higher levels of physical and mental performance (Magyar & Duda, 2000). Finally, maintaining a connection to the team can also help footballers maintain their playing identity and sense of "self" during the recovery phase (Podlog et al., 2014). This can be achieved by inclusion at strategy meetings (analysing match play) and also routine attendance at team meals.

Collaboration between the physical trainer and sports scientist allows the player's physical progress to be monitored and recorded with measurements of propulsive speed, peaks of power and maximum forces, providing objective information about their "readiness" to RTP (Figure 1). Data from global positioning systems (GPS) can monitor the players progress in completing external running loads (maximal speed, ratio of high-speed to low speed running, and distance covered). Other key indicators of anticipated recovery include regaining baseline strength measures, completion of high intensity training comparable to (or even greater than) their anticipated match demands, and demonstration of an appropriate level of football-specific cognitive skills and psychological readiness. Of nutrition relevance, both gelatin and collagen are rich in glycine, proline, hydroxylysine and hydroxyproline; the ingestion of which have been shown to increase collagen

Table I. Practical examples of how sports nutrition and sports psychology may synergistically support the RTP process.

RTP consideration	Nutrition / Psychology interdisciplinary examples
Acute injury	Dietary intake and supplementation plan as an action to achieve sense of control
Body mass management	Routine body composition analysis provides short-and-long term goal setting targets
Inactivity	Meals providing structure to day, avoid episodes of reduced self- worth and binge eating
Social isolation /inclusion	Inclusion of players on team eating occasions and having players complete rehabilitation alongside teammates who are training to maintain sense of identity
Role models	Establish "mentor" programme between injured player and role model who successfully overcame similar injury. Sharing of previous experience of both sports nutrition and sports psychology strategies that "helped" the RTP, to facilitate rehabilitation adherence
Readiness	Nutrition provided prior to exercise and education enhances players understanding of their body to facilitate confidence and enhance psychological readiness (confidence, emotions, risk appraisals, approach-avoidance motivations). Education also serves to inform about best sources of information (avoid unqualified advice from internet/social media)

synthesis as well as improving ligament / tendon mechanics. For musculoskeletal injuries at the myotendinous junction, the ingestion of 15–20 g of hydrolysed collagen, 30 minutes to 1 hour prior to the rehabilitation exercise could be considered to help facilitate healing (Baar, 2017). Although research on psychological readiness is still in its infancy, it may be comprised of several components, including confidence in one's RTP capabilities, emotions (e.g. re-injury apprehensions), risk appraisals, realistic expectations of one's sporting capabilities, and motivation to regain previous performance standards. Assessment of psychological readiness (for example, the Injury Psychological Readiness to Return to Sport measure (Glazer, 2009)) may be used to complement traditional sport medicine assessments focusing exclusively on physical/functional (milestones 3 and 4). As has been recently argued (Ardern, Taylor, Feller, & Webster, 2013; Forsdyke, Smith, Jones, & Gledhill, 2016, 2017), holistic assessment of athletes' readiness to return to sport should include psychological test batteries, working knowledge of and/or observations of athlete behaviours by coaches and members of the sports medicine team, and shared decision-making involving key stakeholders.

Once key performance indicators are achieved the player may be included in part-time training (1–3 d/week). During this period players should be reminded/educated on sports nutrition "basics": commencing exercise with adequate muscle glycogen stores and euhydrated (Williams & Rollo, 2015). Determining the players individual sweat response to training as well as ensuring appropriate carbohydrate availability for the training session may help decrease the risk of re-injury, whilst promoting a good dietary regime.

Following alignment with the interdisciplinary team, the player may commence fully with team activities. A weeks training with the team is usually advised before a player is available for selection. At this stage. coaching staff are encouraged to progressively increase the player's contribution to competition.

Practical considerations. 799067An injury to a player presents an opportunity for both the sport nutritionist and sports psychologist to engage with the player, as well as the sports medicine department. In the authors' experience, sport nutrition and sport psychology support during an injury often results in players, who have been reluctant to work with these disciplines, becoming engaged for the rest of their careers, having experienced support benefit (Table I). Providing the players with appropriate nutrition and psychological "tools" to manage the injury process can also extend and remain relevant to when players are back in full training. Thus, it is advised to continue monitoring the players from a sports nutrition and sports psychology perspective as well as ensure individual programmes are completed, as a preventative measure against re-injury, when the player has returned to first-team training. However, the long-term implications to adhering to these strategies post injury on player performance, well-being, risk of re-injury and club/player finances are yet to be established. Finally, during the injury period it is advisable to reduce, if not eliminate, alcohol intake. Alcohol post exercise has been shown to reduce myofibrillar protein synthesis even if co-ingested with protein (Parr et al., 2014), potentially impairing recovery. Furthermore, alcohol intake increases the risk of excessive energy intake, poor dietary choices and psychological coping strategies failing.

Summary

The interdisciplinary theme is important as it *integrates* concepts, theories, methods, and techniques from two or more disciplines, with the aim of

advancing fundamental knowledge and the solving of 'problems' and societal challenges in and via the sport and exercise sciences. An interdisciplinary approach is understood to achieve outcomes that could not be achieved within the framework of a single discipline. The current article highlights opportunities for incorporating sports psychology and sports nutrition into phased medical approaches to injury rehabilitation and RTP. However, it remains uncertain whether or how adoption of an interdisciplinary approach speeds or facilitates a more effective and/or comprehensive recovery in comparison to single discipline approaches. It is the authors opinion that such an interdisciplinary approach will indeed be advantageous for the RTP process – but this paper is also a "call to action" for additional research. This research, and associated applied practice efforts, is needed to examine the value of an interdisciplinary approach in facilitating player functional and RTP outcomes.

Disclosure statement

IR and JC are employees of the Gatorade Sports Science Institute, a division of PepsiCo, Incorporated. The views expressed in this article are those of the authors and do not necessarily reflect the position or policy of PepsiCo, Incorporated.(s).

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